

P100 Unit Diagnostic Report

Modem Failure - Sierra Wireless EM7565 Absent from USB Bus

Node (planename)	N120AV
Airline	AVA (Avianca)
Serial Number	W03021890023
Diagnostic Date	August 20, 2026
Diagnosis	Hardware failure - modem no longer enumerating on USB bus
Modem State	mmcli -L: 'No modems were found'; not present in lsusb
Prior History	Previously diagnosed with a damaged SIM interface controller (traced to an interrupted firmware upgrade)
Rest of Unit	USB bus otherwise healthy - Teensy Board and other USB devices detected normally

Diagnostic Summary

Unit N120AV was previously diagnosed with a damaged SIM interface controller on its Sierra Wireless EM7565 modem, originating from a firmware upgrade that was interrupted during a prior repair attempt. At that time, the modem still enumerated over USB and reported failed reason: `sim-missing`. On this follow-up diagnostic session, the modem's condition has progressed further: it no longer enumerates on the USB bus at all. ModemManager reports no modem present, and the modem does not appear anywhere in the system's USB device list, with no trace of it in the kernel logs. The rest of the unit's USB bus is confirmed healthy - other internal USB devices (including the Teensy Board) are detected and enumerate normally, isolating this as a modem-specific hardware failure rather than a broader USB controller issue.

Diagnostic Steps Performed (in order)

1. Checked for the modem via ModemManager (`mmcli -L`): result 'No modems were found', compared to the previously documented state where the modem was detected but reported `sim-missing`.
2. Checked for the modem's expected system symlinks (`/dev/modemPPP`, `/dev/modemAT`): neither exists.
3. Searched the USB device list specifically for the Sierra Wireless modem (`lsusb | grep -i sierra`) and searched kernel logs for any related activity (`dmesg | grep -iE 'sierra|1199'`): no results in either case - no trace of the modem hardware.
4. Reviewed the complete USB device list (`lsusb`) to determine whether this was an isolated modem failure or a broader USB bus issue: all other expected devices are present and enumerating normally, including the Teensy Board (16c0:0483), the FTDI serial converters, and the internal USB hub - confirming the failure is isolated to the modem.

Conclusion: this diagnostic confirms and extends the prior finding of a hardware failure in the Sierra Wireless EM7565 modem's SIM interface controller. The modem's condition has since progressed from being detected but unable to read the SIM, to no longer enumerating on the USB bus at all - consistent with continued degradation of already-damaged hardware. All other USB subsystems on the unit remain healthy. Physical replacement of the modem is recommended; this unit was previously flagged for the same action and remains pending physical replacement.

Appendix: Technical Evidence (logs and command output)

Unit Identification

N120AV
AVA
Serial Number: W03021890023
UUID: 03000200-0400-0500-0006-000700080009
Wake-up Type: Power Switch
SKU Number: SGP100-713020-00;M0-000;W0-100;B0-15

ModemManager Device List (mmcli -L)

No modems were found

System Symlinks

```
$ ll /dev/modemPPP /dev/modemAT
ls: cannot access '/dev/modemPPP': No such file or directory
ls: cannot access '/dev/modemAT': No such file or directory
```

Modem-Specific USB / Kernel Log Search

```
$ lsusb | grep -i sierra
(no output)
```

```
$ dmesg | grep -iE "sierra|1199"
(no output)
```

Note: no trace of the Sierra Wireless modem (USB vendor ID 1199) anywhere in the USB device list or kernel boot log.

Full USB Device List (lsusb)

```
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 006: ID 16c0:0483 Van Ooijen Technische Informatica Teensyduino Serial
Bus 001 Device 005: ID 0403:6015 Future Technology Devices International, Ltd Bridge(I2C/SPI/UART/FIFO)
Bus 001 Device 004: ID 05e3:0608 Genesys Logic, Inc. Hub
Bus 001 Device 003: ID 8087:0025 Intel Corp.
Bus 001 Device 002: ID 0403:6010 Future Technology Devices International, Ltd FT2232C/D/H Dual UART/FIFO IC
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

Note: the Teensy Board and all other expected internal USB devices are present and enumerating normally. Only the Sierra Wireless modem (previously present at USB vendor ID 1199) is absent - confirming the failure is isolated to the modem and not a broader USB bus issue.

Prior Diagnostic History (reference)

This unit was previously diagnosed with a damaged SIM interface controller, originating from a firmware upgrade that was interrupted during a prior repair attempt. At that time:

failed reason: sim-missing

sim slot paths: slot 1: none (active)

The modem was still detected by ModemManager at that time. It is no longer detected at all as of this diagnostic session, indicating further degradation of the same underlying hardware fault.